This is 2nd lecture note on Cherenkov effect. Here we work in frame K (Lab frame) from the very beginning.

$$\vec{\nabla}\cdot(\varepsilon\vec{E}) = \rho(\vec{r}) = q\,\delta(\vec{r}-\vec{v}t) \quad , \vec{\nabla}\cdot\vec{B}=0$$

$$\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t} \quad , \vec{\nabla}\times\vec{H} = \frac{\partial\vec{D}}{\partial t}+\vec{j} \Rightarrow \vec{B}=\mu\vec{H}\to\mu_0\vec{H}$$

$$\Rightarrow \vec{\nabla}\times\vec{B} = \mu_0\vec{j} + \mu_0\frac{\partial}{\partial t}(\varepsilon\vec{E}) \quad , \vec{j}=q\vec{v}\,\delta(\vec{r}-\vec{v}t)$$

Let us introduce standard potentials $\varphi, \vec{A}$: $\vec{E} = -\vec{\nabla}\varphi - \frac{\partial\vec{A}}{\partial t}$, $\vec{B}=\vec{\nabla}\times\vec{A}$

Then Faraday's law and $\vec{\nabla}\cdot\vec{B}=0$ are satisfied.

$$\Rightarrow \vec{\nabla}\times(\vec{\nabla}\times\vec{A}) = \vec{\nabla}(\vec{\nabla}\cdot\vec{A}) - \nabla^2\vec{A} = \mu_0\frac{\partial}{\partial t}\left(-\varepsilon\vec{\nabla}\varphi - \varepsilon\frac{\partial\vec{A}}{\partial t}\right) + \mu_0\vec{j}$$

$$\Rightarrow \frac{\partial}{\partial t}\mu_0\varepsilon\frac{\partial\vec{A}}{\partial t} - \nabla^2\vec{A} = \mu_0\vec{j} - \vec{\nabla}\underbrace{\left(\frac{\partial}{\partial t}\mu_0\varepsilon\varphi + \vec{\nabla}\cdot\vec{A}\right)}_{\text{require this}=0} \quad (1)$$

require this $=0 \Rightarrow$ this is just Lorentz gauge generalization to the medium.

$$\Rightarrow \text{Gauss' law} \quad -\vec{\nabla}\cdot\left(\varepsilon\vec{\nabla}\varphi + \varepsilon\frac{\partial\vec{A}}{\partial t}\right) = -\varepsilon\left(\nabla^2\varphi + \frac{\partial}{\partial t}\vec{\nabla}\cdot\vec{A}\right) = +\varepsilon\left\{\frac{\partial^2}{\partial t^2}(\varepsilon\mu\varphi) - \nabla^2\varphi\right\}$$

$$\vec{\nabla}\cdot\vec{A} = -\frac{\partial}{\partial t}\varepsilon\mu\varphi$$

$$\Rightarrow \varepsilon\left(\frac{\partial^2}{\partial t^2}(\varepsilon\mu_0\varphi) - \nabla^2\varphi\right) = \rho(\vec{r},t). \quad (2)$$

So we obtained wave equations (1) and (2) for $\vec{A}$ and φ in medium.

We solve them by F.T. Use $\delta(\vec{r}-\vec{v}t) = \int\frac{d\vec{k}}{(2\pi)^3}e^{i\vec{k}(\vec{r}-\vec{v}t)} =$

$$= \int\frac{d\vec{k}}{(2\pi)^3}\int\frac{d\omega}{2\pi}\,e^{i\vec{k}\cdot\vec{r}-i\omega t}\,2\pi\,\delta(\omega-\vec{k}\cdot\vec{v})$$

Therefore R.H.S. of these equations is Fourier transformed as follows:

$$\delta(\vec{r}-\vec{v}t) \to 2\pi\,\delta(\omega-\vec{v}\cdot\vec{k}).$$

Thus
$$\begin{cases} \varepsilon(k^2-\varepsilon\mu_0\omega^2)\,\varphi(\vec{k},\omega) = 2\pi q\,\delta(\omega-\vec{v}\cdot\vec{k}) \\ (k^2-\varepsilon\mu_0\omega^2)\,\vec{A}(\vec{k},\omega) = 2\pi\mu_0 q\,\vec{v}\,\delta(\omega-\vec{v}\cdot\vec{k}) \end{cases}$$

$$\varphi(\vec{r},t) = \int\frac{d\vec{k}}{(2\pi)^3}\frac{d\omega}{2\pi}\frac{2\pi q\,\delta(\omega-\vec{v}\cdot\vec{k})}{\varepsilon[k^2-\varepsilon\mu_0\omega^2]}e^{i\vec{k}\cdot\vec{r}-i\omega t}$$

$$= \int\frac{d\vec{k}}{(2\pi)^3}\frac{q\,e^{i\vec{k}\cdot(\vec{r}-\vec{v}t)}}{\varepsilon[k^2-\varepsilon\mu_0(\vec{v}\cdot\vec{k})^2]}$$

$$\vec{A}(\vec{r},t) = \int\frac{d\vec{k}}{(2\pi)^3}\frac{\mu_0 q\,\vec{v}\,e^{i\vec{k}\cdot(\vec{r}-\vec{v}t)}}{k^2-\varepsilon\mu_0(\vec{v}\cdot\vec{k})^2} = \varepsilon\mu_0\vec{v}\,\varphi(\vec{r},t).$$

As before, $\vec{v}\cdot\vec{k} = v k_z$, $\varepsilon\mu_0 = \frac{\varepsilon\mu_0}{\varepsilon_0\mu_0 c^2} = \frac{n^2}{c^2}$

$\Rightarrow$ denominator has form $k^2 - \frac{n^2}{c^2} v^2 k_z^2 = k_\perp^2 + (1-n^2\beta^2) k_z^2$

For Cherenkov effect $n\beta = \frac{v}{c/n} > 1 \Rightarrow$ re-write

$$\varphi(r,t) = -\frac{q}{\varepsilon}\int \frac{d^2\vec{k}_\perp}{(2\pi)^2} e^{i\vec{k}_\perp\cdot\vec{r}} \int_{-\infty}^{\infty} \frac{dk_z}{2\pi} \frac{e^{ik_z(z-vt)}}{(n^2\beta^2-1)k_z^2 - k_\perp^2}$$

k_z-integral has poles at $k_z = \pm \frac{k_\perp}{\sqrt{n^2\beta^2-1}}$ (note $k_\perp = \sqrt{k_x^2+k_y^2} > 0$)

These poles are <u>not</u> on real axis, actually, because $n = n' + i n''$ in fact has infinitesimal imaginary part $n''(\omega)$. Moreover $\begin{cases} n''(\omega) > 0 \text{ for } \omega > 0 \\ n''(\omega) < 0 \text{ for } \omega < 0. \end{cases}$

Therefore need to expand $(n^2\beta^2-1)^{-1/2} = \left[\beta^2(n'^2 + i\,2n'n'') - 1\right]^{-1/2} =$

$$= \left[\beta^2 n'^2 - 1 + i\,2\beta^2 n' n''\right]^{-1/2} = (\beta^2 n'^2 - 1)^{-1/2}\left(1 - i\frac{\beta^2 n' n''}{\beta^2 n'^2 - 1}\right)$$

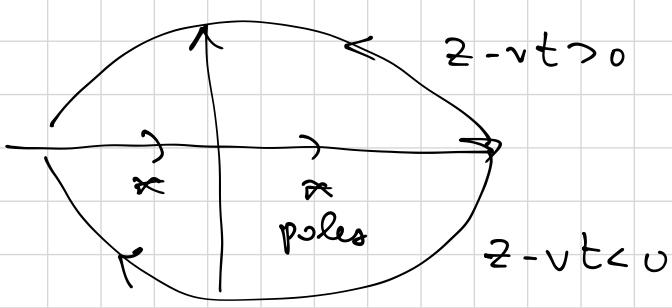
$\Rightarrow$ poles are at $k_z^{(\pm)} = \pm \frac{k_\perp}{\sqrt{\beta^2 n'^2 - 1}}\left(1 - \frac{i\beta^2 n' n''}{\beta^2 n'^2 - 1}\right)$.

Here $n''(\omega) = n''(v k_z)$ due to the $\delta(\omega - \vec{v}\cdot\vec{k})$.

Hence for $\oplus$ sign, when $k_z^{(+)} > 0$, $n''(v k_z) > 0 \Rightarrow \mathrm{Im}(k_z^{(+)}) < 0$.

For $\ominus$ sign, when $\mathrm{Re}(k_z^-) < 0$, $n''(v k_z) < 0 \Rightarrow \mathrm{Im}(k_z^{(-)}) < 0$ again.

Thus both poles are in $\mathrm{Im}(k_z) < 0$ half-plane!



Therefore for $z - vt > 0$ the integral is 0.
(need to close contour in $\mathrm{Im}(k_z) > 0$ half-plane, where $e^{i(z-vt)[k_z' + i k_z'']} \sim e^{-k_z''(z-vt)}$ is converging. But there are no poles!)

For $z - vt < 0$ both poles contribute $\tilde{z} = (z-vt)/\sqrt{n^2\beta^2-1}$

$$\varphi(\vec{r},t) = -\frac{q}{2\pi\varepsilon(\beta^2 n^2 - 1)}\int dk_\perp k_\perp \int_0^{2\pi}\frac{d\varphi}{2\pi} e^{ik_z\rho\cos\varphi}\left(\frac{-2\pi i}{2\pi}\right)\frac{2i\sin(\tilde{z}k_\perp)}{2k_\perp}\sqrt{n^2\beta^2-1}\, e^{-\alpha k_\perp}$$

$$= \frac{-q}{2\pi\varepsilon\sqrt{\beta^2 n^2 - 1}}\int_0^{2\pi}\frac{d\varphi}{2\pi}\frac{1}{2i}\left(\frac{1}{\alpha - i(\rho\cos\varphi + \tilde{z})} - \frac{1}{\alpha - (\rho\cos\varphi - \tilde{z})}\right)$$

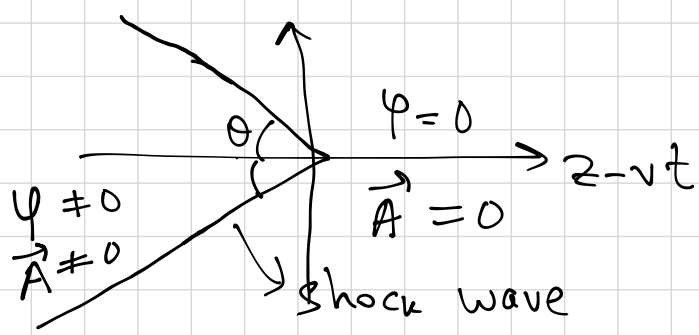
convergence factor. $\alpha = +0$

Using Sohotsky's identity $\frac{1}{x \pm i0} = P\frac{1}{x} \mp i\pi\delta(x)$ and $-\tilde{z} = \rho a$:

$$\varphi(\vec{r},t) = \frac{q}{2\pi\varepsilon\rho\sqrt{\beta^2 n^2 - 1}} \underbrace{P.\int_0^{\pi} \frac{d\varphi}{\pi} \frac{1}{a - \cos\varphi}}_{\frac{\theta(a-1)}{\sqrt{a^2-1}}} = \frac{q}{2\pi\varepsilon\sqrt{\beta^2 n^2 - 1}} \frac{1}{\sqrt{\tilde{z}^2 - \rho^2}}$$

for $-\tilde{z} > \rho$.

$$\varphi(\vec{r},t) = \begin{cases} \frac{q}{2\pi\varepsilon} \frac{1}{\sqrt{(z-vt)^2 - (\beta^2 n^2 - 1)(x^2+y^2)}} & \text{inside the cone} \\ 0 & \text{outside} \end{cases}$$



$$\sin\theta = \frac{1}{\beta n}$$

$$\vec{A} = \mu_0 \varepsilon \vec{v} \varphi = \frac{n^2 \vec{v}}{c^2} \varphi$$

$$\vec{E} = -\vec{\nabla}\varphi - \frac{\partial \vec{A}}{\partial t} = -\vec{\nabla}\varphi - \frac{n^2 \vec{v}}{c^2}\frac{\partial \varphi}{\partial t} \quad \text{but} \quad \frac{\partial \varphi}{\partial t} = v\frac{\partial \varphi}{\partial (vt)} = -v\frac{\partial \varphi}{\partial z}$$

$$\Rightarrow \vec{E} = -\partial_x \varphi\, \hat{x} - \partial_y \varphi\, \hat{y} + (n^2\beta^2 - 1)\partial_z \varphi\, \hat{z}.$$

$$\vec{B} = \vec{\nabla} \times \vec{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ 0 & 0 & \frac{n^2 v}{c^2}\varphi \end{vmatrix} = \frac{n^2 v}{c^2}\left(\partial_y \varphi\, \hat{x} - \partial_x \varphi\, \hat{y}\right).$$

(it is convenient to use cylindrical coordinates for further calculations.)